9.8 Nobody Gets Their Own Shirt

WORKSHEET

The question

STEP II 2016, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q12

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Pause and try Try $n = 3$: list all six orders of cards $1, 2, 3$ and count how many have no card in its own place. Then try $n = 4$, if you are patient (there are 24 orders).

STAGE 1 Inclusion–exclusion for three events

Ask yourself How can you write $A \cup B \cup C$ as a union of two events, so that the two-event formula applies?

STAGE 2 The probabilities

Ask yourself If card i is in position i, how many ways are there to arrange the rest? And if cards i and j are both fixed?

STAGE 3 Adding it up

Ask yourself In the inclusion–exclusion formula for $\mathrm{P}(E_1 \cup \cdots \cup E_n)$, how many terms are there involving intersections of k events, and what is their total?

STAGE 4 Exactly one

Ask yourself For exactly one card to be in place, which card is it, and what must the others do?

Try this next

1 **Shirts.** (From STEP III, 2004.) A team of m players put on m numbered shirts at random. Let $C_i = 1$ if player i wears shirt i, and 0 otherwise. Find $\mathrm{E}(C_1)$, $\mathrm{Var}(C_1)$ and $\mathrm{Cov}(C_1, C_2)$, and show that $N = C_1 + \cdots + C_m$ has $\mathrm{E}(N) = \mathrm{Var}(N) = 1$. For $m = 4$, compare $\mathrm{P}(N = 0)$ with the Poisson approximation e^{-1}.

2 **A recurrence.** Let D_n be the number of derangements of n objects. By considering where card 1 goes, and whether the card in that position goes to position 1, show that $D_n = (n-1)(D_{n-1} + D_{n-2})$. Check it against $D_3 = 2$, $D_4 = 9$.

3 **Exactly k.** Show that the probability that exactly k cards are in their own places is

$\frac{1}{k!}d_{n-k}$, and that it tends to $\frac{e^{-1}}{k!}$.

4 **Coprime integers.** Use inclusion–exclusion to count the integers from 1 to 30 divisible by none of 2, 3 and 5. How does your answer relate to $30(1-\frac{1}{2})(1-\frac{1}{3})(1-\frac{1}{5})$?

Further reading

- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press). A first-year text that covers inclusion–exclusion and the matching problem.
- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter IV. A classic, with many variations on the matching problem.
- **Miklós Bóna, *A Walk Through Combinatorics*** (World Scientific), on inclusion–exclusion.

The full guided solution, and what the question leads to at university, is in Section 9.8:
https://beyondstep.keithmcnulty.org/ch09/derangements.html